| Problem Chosen | 2020 MCM/ICM Summary Sheet | Team Control Number |
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| A | | 2013644 |

# Saving Scotland-based Fishing Companies in Danger: Go North as the Fishes Do


As two main species of Scottish fishery harvest, Scottish herring and mackerel are moving north away from the main coast of Scotland due to the increasing ocean temperature, seriously influencing the earning of the Scotland-based fishing companies, small companies in particular. It is of great importance to investigate on the current situation of the problem and to propose possible solutions.

In this paper, we build several mathematical models to study how Scottish herring and mackerel will migrant in the next thirty years and seek practical and economical strategies to save the Scotland-based fishing companies which are in danger of going bankrupt.

**Firstly**, we try to figure out the warming pattern of global sea surface temperature (SST). Using the data collected, we use Python to visualize the isothermal map and use least square method to build the **Ocean Warming Model**, which predicts the warming pattern of global SST in the future based on the warming pattern in the past. The generated polynomial model fits the given data sets well, and the result of the prediction can be utilized to approximate the differences in temperature between a given year and the year of 2020.

**Secondly**, we investigate the **temperature preference** of the Scottish herring and mackerel, which can be inferred by looking at their population distribution under different temperatures. By fitting the population distribution into a normal distribution, it is found that the optimum temperature of herring is 5℃ and that of mackerel is 10.2℃.

**Thirdly**, we build the **Habitat Suitability Evaluation Model** by introducing the concept of **suitability index**, the extent to which an environment is suitable for a creature to live in. In this paper, we assume that the suitability of habitat is only related to temperature.

**Fourthly**, we simulate the migration of Scottish herring and mackerel by implementing **Cellular Automation Algorithm**. It is found that both herring and mackerel are moving north-west and north-east away from Scotland, and it takes only 15 to 20 years for herring and 55 to 80 years for mackerel to migrate too far away for small fishing companies to harvest.

**Fifthly**, we build the **Small Company's Payoff Model** to estimate the total profit a fishery company can make per year, depending on the amount of fish harvested. We propose strategies such as relocating the company's assets to ports where both fish move towards, equipping vessels with cold storage facilities, renting other company's vessels and changing the target species of harvesting. Then we adjust the model based on the change of strategies.

**Finally**, we discussed issues when some proportion of the fishery moves into the territorial waters of another country.



**Key words:** Warming Pattern of the Ocean, Habitat Suitability Index, Cellular Automaton, Small Company's Payoff Model

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# I Introduction

## 1.1 Problem Restatement

Unlike human for whom the global warming does not lead to large-scale migration, certain ocean-dwelling species display gradual but distinct population migration over the last few years. With the original habitats having a temperature that is increasing too fast for their suitable living and reproduction, Scottish herring and mackerel, like the lobster in Maine, are two potential species to become new immigrants to the oceans with lower temperatures. As two fish species that take up a significant economic contribution of the Scottish fishing industry, their migration will certainly affect the survival of smaller Scotland-based fishing companies who lacks financial support to equip their vessels with on-board refrigeration. As the herring and mackerel inhabit farther and farther from the coast, if taking no actions, the small companies seem to be doomed to either catch little fishes near the coastline or come back to Scotland with all the harvests inedible.

As consultants hired by the Scottish North Atlantic Fishery Management Consortium, we aim to analyze the effects of the warming waters on the operation of the small fishing companies and provide them with practical advice.

To be specific, there are three major tasks to accomplish:

1. According to the current global warming pattern and its influence on the warming of the seawater, we should build a model to predict the ocean temperature at a specific place in the Scottish North Atlantic at a specific time.

2. Though the increase in ocean temperature is the driving source of the migration of the fish, their pattern of migration does not simply follow the change of the isothermals. Therefore, we should treat it as a problem of evolution that is largely affected by ocean warming, and we should investigate the process of migration based on different warming speeds.

3. Based on the built models, determine when the herring and mackerel will be too far away from the small fishing companies. Given the trend of the market quotation, we should provide the companies with some corresponding suggestions on their operations.

## 1.2 Problem Analysis

As an ecological “Game of Life” problem, the migration of fish is essentially

a dynamic process which emphasizes the interaction between the individuals as well as the interaction between them and the environment. Therefore, we intend to apply Cellular Automation to simulate the process of migration. Before that, we need to build some models to investigate the relations between some important variables.

As emphasized in the problem, the migration is essentially caused by the change of temperature. Therefore, the first model we build is the **Ocean Warming Model**, which is intended to investigate the warming pattern of the ocean. Since it is very complicated to analyze the heat transfer process between the air and the sea surface as a physicist or a chemist, we search for abundant data of the sea surface temperature (SST) in order to predict the warming pattern in the future based on the warming pattern in the past. We use the least square regression to predict a polynomial model to fit the given data sets, and the result of the prediction can be utilized to approximate the differences in temperature between a given year and the year of 2020.

Next, we intend to build a **Habitat Suitability Evaluation Model** that determines the extent to which an environment is suitable for a type of fish to live in. To achieve this, we investigate the temperature preference of the Scottish herring and mackerel, which can be inferred by looking at their population distribution under different temperatures. By sampling survey, the population distribution is proven to agree with a normal distribution, and thus the temperature preference of the fish should also be normally distributed.

After that, with each temperature corresponding to a suitability index, the suitability of each location on the map is calculated. In the **Cellular Automation Algorithm**, the Scottish herring and mackerel are initially populated on the map following their temperature preference, and their migration is implemented by finding the biggest suitability index in the adjacent area. By always moving to the place with the highest suitability index in the area, the simulated fish migrates continuously. According to the result of the simulation, herring can be too far away for the Scotland-based fishing companies to harvest in 15 years.

Last, we build the **Small Company's Payoff Model** to figure out whether there is a way to save the small companies from going broke. Following the basic logic of "letting the profit exceed the cost", we consider the cost and the earnings in as many aspects as possible (e.g., power cost, labor cost, cost of removal and cost of refrigerator) and verify this model by simulation.

# II Assumptions and Justifications

Aiming to simplify the problem and to hold the correctness of the results at the same time, we make the following assumptions in our models, with the

rationality of each of them explained.

1. During the time scale discussed in this paper, the earth's ecological environment keeps stable. That is, we don't consider the incidents that may cause drastic climate changes such as oceanic earthquake or Oil Spill, which have small probabilities to happen.

2. Scottish herring and mackerel stick to their biological habit of clustering. Therefore, the migration of one unit of fish (5,000 tons) can be seen as the moving of one or some clusters of electron cloud.

3. The overall population of the herring and mackerel living in Scottish North Atlantic has reached an equilibrium, and their population won't have considerable change because of migration.

4. The temperature in different small area increases in the same way. That is, the differences of temperature between different areas will not change through time.

5. Temperature is the only variable that determines the pattern of migration of the Scottish herring and mackerel. Admittedly, there are plenty of other influential, and we only consider the factor of temperature in this paper because of the context of the problem.

# III Notations

For concision, we use the following symbols in our models, as listed in Table 1.

Table 1. Symbols, Definitions and Units

| Symbol | Definition | Unit |
|---|---|---|
| T | sea surface temperature (SST) | ℃ |
| ΔT | change of SST between two years | ℃ |
| i | latitude | / |
| j | longitude | / |
| t | AD | year |
| $H_m$ | guarantee time for the fish to be eaten raw | h |
| $S_m$ | typical headway of a vessel | km/h |
| $D_m$ | largest distance a vessel can travel to keep the caught fish fresh | km |

| | | |
|---|---|---|
| $I_s$ | suitability index: the extent of which a habitat is suitable for a creature to live in | / |
| $d_t$ | population density at temperature t | $\text{cell}^{-1}$ |
| X | index of harvested Scottish herring | / |
| Y | index of harvested Scottish mackerel | / |
| F | final profit made by each small fishery company per year | £ |

# IV Game of Life: Location of Scottish Herring and Mackerel in 50 Years

## 4.1 The Warming Pattern of Scottish North Atlantic

As the driving force of the migration of the fishes, the warming pattern of the Scottish North Atlantic is at the core of our model. There has been considerable analytical model for understanding the sea surface temperature patterns under the global warming, and their work are both theoretically and experiment-ally tenable. However, due to the abundant data of the sea surface temperature (SST), in this paper we employ the data from *MOHC observations datasets* [1] and *NOAA* [2], which are utilized to generate the formula of the change of global SST $\Delta T$ with respect to a year.

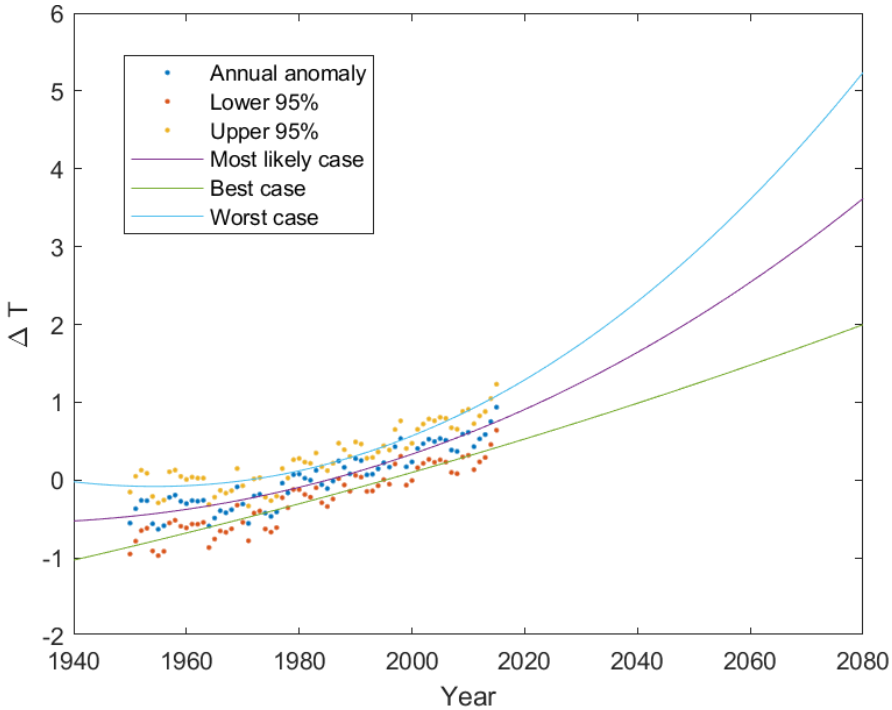
### 4.1.1 $\Delta T$ with Respect to Year

Based on the data of the average global SST from 1950 to 2015, we use least square regression to predict a polynomial model to fit the given data sets.

We assume that the temperature in different small area increases in the same way (i.e., the differences of temperature between different areas will not change), and we initialize the temperature in each small area with the ocean temperature data of January 2020. We use least square regression to predict a polynomial model to fit the given data sets. The result of the prediction can be utilized to approximate the differences of temperature $\Delta T$ between a given year $t$ and the year of 2020 (Figure 1), and the function of change of temperature can be written as:

$$\Delta T = 4.0393 \times 10^{-11}\, y^4 - 1.5477 \times 10^{-6}\, y^3 + 1.4677 \times 10^{-4}\, y^2 + 2.5780 \times 10^{-3} y - 0.9079 \tag{1}$$

Figure 1. Annual anomaly in 2020 and the Predicted value in 2070




Similarly, we use the data of the lower 95% confidence interval and the upper 95% confidence interval to generate the following formulas

$$\Delta T_L = 3.3630 \times 10^{-12}\, y^4 - 7.8350 \times 10^{-9}\, y^3 + 1.3635 \times 10^{-6}\, y^2 + 1.7553 \times 10^{-3} y - 0.5280 \tag{2}$$

$$\Delta T_U = 7.7423 \times 10^{-11}\, y^4 - 3.0171 \times 10^{-7}\, y^3 + 2.9217 \times 10^{-4}\, y^2 + 3.4007 \times 10^{-3} y - 1.2878 \tag{3}$$

where $\Delta T_L$ represents the smallest $\Delta T$ between year $t$ and year 2020 that holds a considerable probability, and $\Delta T_U$ represents the largest $\Delta T$ between year $t$ and year 2020 that holds a considerable probability.

### 4.1.2 Visualize the SST Pattern of the Scottish North Atlantic in January

According to the Scottish government, herring and mackerel are caught the largest in quantity in January [3]. Therefore, in this paper we focus on the ocean warming pattern and the distribution of the fish in January each year, and we believe that January is the month that the fishing companies in Scotland care about the most.

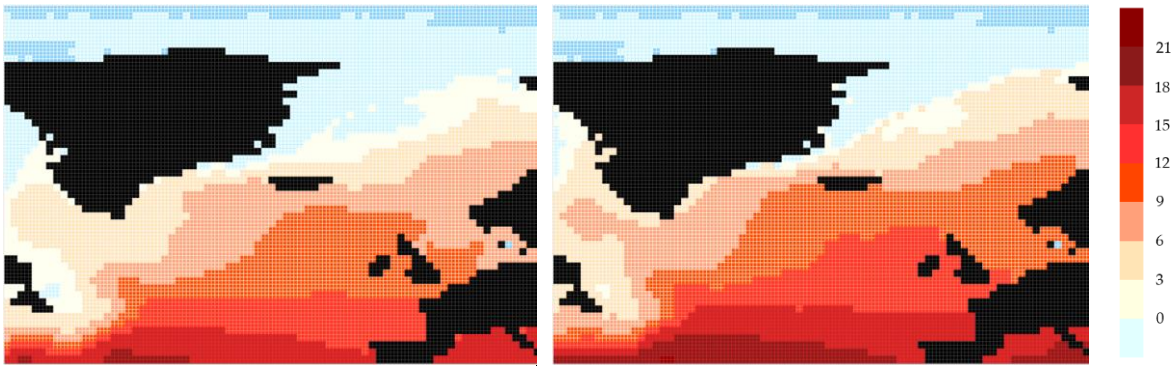
Having had the relation between the SST of 2020 and that of any given year,

we can approximate the SST of any place at any time as long as the data of SST of 2020 is available.

We acquire historical global SSTs from the mid-19th century to present from database *Hadisst* [1] and apply (1) on these data, which helps predict the SST at any specific location of any given year. In order to have a big idea of the sea surface temperature, we write a program to visualize the SST pattern of the Scottish North Atlantic near Scotland in January (the code is attached in **Appendix B**). The range of latitude of the canvas is [40°N, 90°N] and the range of longitude is [60°W, 15°E]. The canvas is divided into 100 × 150 small cells, with each cell represent the area of 0.5°x0.5°.

In the following generated figure, it is shown that as time goes by, the isothermals are moving towards the north-west gradually.

Figure 2. SST of Scottish North Atlantic in 2020 (left) and 2070 (right)



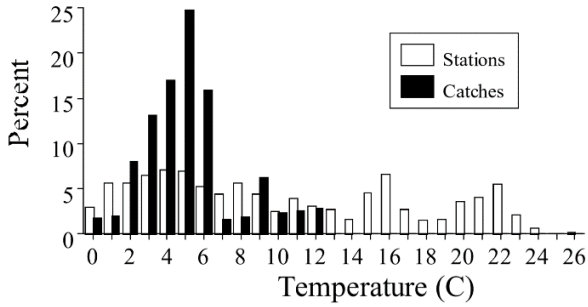

## 4.2 Habitat Suitability Evaluation Model

### 4.2.1 The Temperature Preference of Herring and Mackerel

In order to find out the temperature preference of Scottish herring and Mackerel, we need to know their distribution under different temperature. In this paper, we use the relative number of Scottish herring (or Mackerel) caught in different thermal environment to represent the population of that type of fish under the corresponding temperature, which generates a population distribution function with respect to temperature. Without loss of generality, we assume the population distribution agrees with a normal distribution.

From *Essential Fish Habitat Source Document* published by NOAA, we find the required data on Scottish herring's life history and habitat characteristics [4]. In terms of mackerel, the information is collected from one website called *Fishbase* [5].

Figure 3. Relative Number of Caught Herring Implies Population Distribution




The figure above shows the mean water temperature at stations where adult Scottish herring were collected (solid bars) and at all stations (open bars) during Hudson-Raritan trawl surveys [4]. The percentage of Scottish herring caught under different temperature is roughly normal distributed. By doing simulation and after adjustment, we derive the probability density of the population distribution of herring with respect to temperature $f_1(T)$ as follows:

$$f_2(T) = \frac{1}{2.3\sqrt{2\pi}} e^{-\frac{(T-5)^2}{2\times 2.3^2}} \tag{4}$$

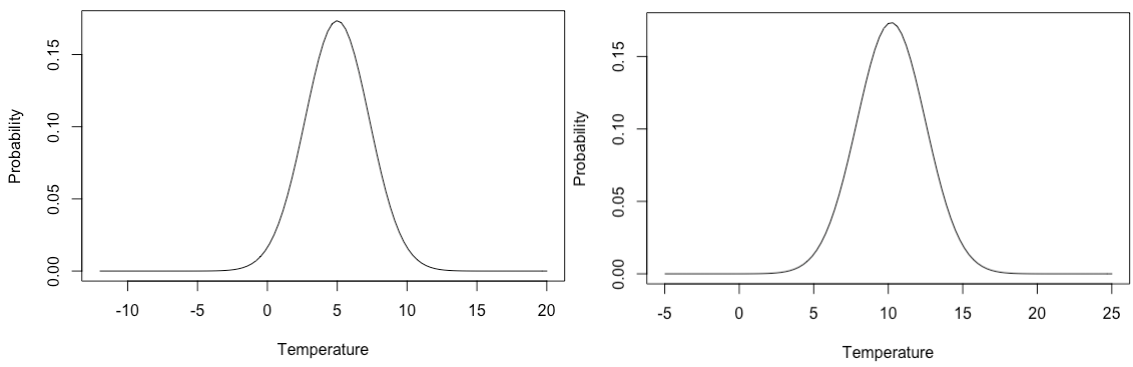
This normal distribution has a mean of 5 ℃ and a standard deviation of 2.3 ℃, meaning that the most preferable temperature for adult Scottish herring is 5 ℃, with a variability of 2.3 ℃.

Similarly, the probability density of the population distribution of mackerel with respect to temperature $f_2(T)$ is determined to be

$$f_2(T) = \frac{1}{2.3\sqrt{2\pi}} e^{-\frac{(T-10.2)^2}{2\times 2.3^2}} \tag{5}$$

This normal distribution has a mean of 10.2 ℃ and a standard deviation of 2.3 ℃, meaning that the most preferable temperature for mackerel is 10.2 ℃, with a variability of 2.3 ℃. T

Figure 4. Probability Density of the Population Distribution of two types of fish with respect to Temperature (left: herring, right: mackerel)



The plots of the above two distribution are shown in Figure 3, from which it is seen that though the most preferable temperature for Scottish herring and mackerel are different, the sensibility of herring and mackerel to the change of temperature is almost the same.

### 4.2.2 Habitat Suitability Index

The habitat suitability index $I_s$ is used to calculate the extent to which a given area is suitable for the Scottish herring or mackerel to live. At the beginning, we intend to consider the effect of two variables, water temperature and plankton concentration, on $I_s$. However, given that plankton concentration is also affected by water temperature as well as the fact that in this "moving North" problem the effects of temperature change is the most essential topic to investigate with, we simplify the model and assume that $I_s$ is only related to water temperature. Further, we assume that $I_s$ is proportional to the probability density of the population distribution of two types of fish with respect to temperature.

From model 4.2.1 we already know the temperature preference of both Scottish herring and mackerel and the probability density of the population distribution of them with respect to temperature.

For herring, we set a place at $T = 5°\mathrm{C}$ has $I_s$ equals to 1, which is the largest $I_s$ possible. In other words, the area of ocean at $5°\mathrm{C}$ cannot be more suitable for the herring to live in (regardless of other factors). Given that $f_1(5) \approx 0.16$, it should be multiplied with 6.25 so as to make it equal to 1. Therefore, the formula for calculating the $I_s$ for herring of an area $I_{s1}$ is

$$\rho_1 = 6.25 \times f_1(T) = \frac{6.25}{2.3\sqrt{2\pi}} e^{-\frac{(T-5)^2}{2\times 2.3^2}} \tag{6}$$

Similarly, the formula for calculating the $I_s$ for mackerel of an area $I_{s2}$ is

$$\rho_2 = 6.25 \times f_2(T) = \frac{6.25}{2.3\sqrt{2\pi}} e^{-\frac{(T-10.2)^2}{2\times 2.3^2}} \tag{7}$$

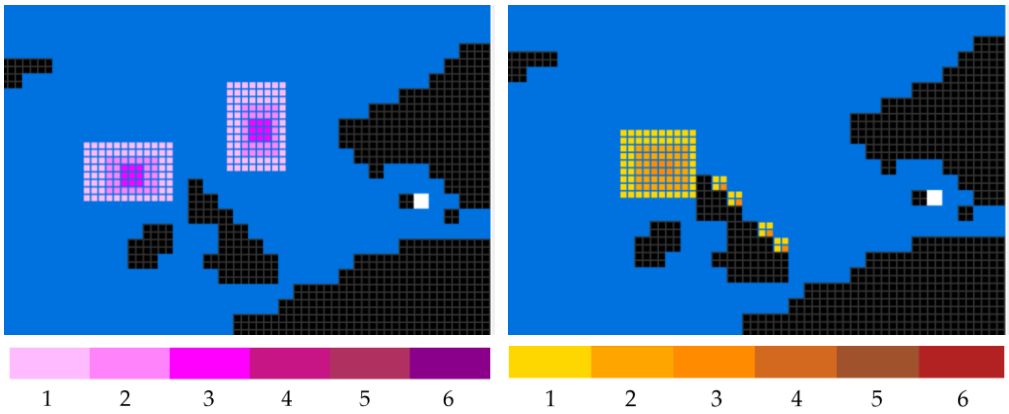
## 4.3 A Cellular Automaton-based Simulation of Migration

### 4.3.1 Initialization: Populate the Fish

As discussed in 4.2.1, the probability density of the population distribution of mackerel with respect to temperature is a normal distribution, therefore, the cell with the temperature that suits the herring or mackerel's needs best will have a larger population density. Since the optimum temperature for herring is 5 °C

and that for mackerel is 10.2 ℃, we locate the fish at the cells with their corresponding optimum temperature, with the center of the population having a larger density than the edges. In Figure 4, 6 different colors are plotted for each type of fish to indicate the different population density of different cells.

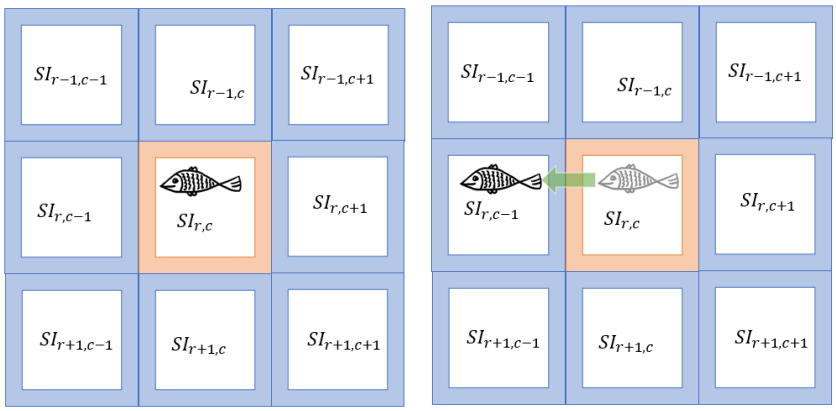
Figure 5. Location of the Herring Colony (purple) and the Mackerel Colony (yellow) in Jan. 2020




### 4.3.2 The Migration of the Fish

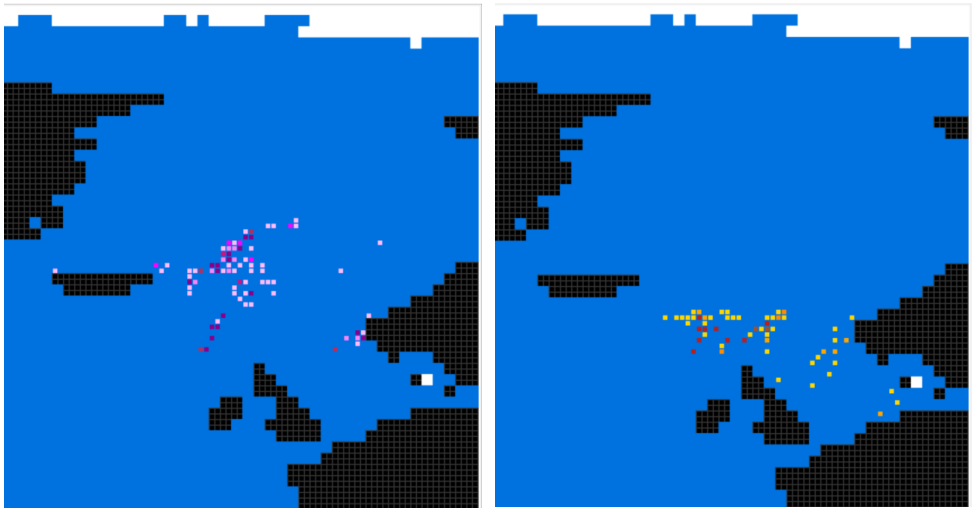
With the assistance of the Habitat Suitability Evaluation Model that is only dependent on temperature, the suitability index $I_s$ of any given point becomes easily accessible. Assume that the decision of which area to move to only depends on the suitability index and the fish can move 0.5° (approximately 55 kilometers) per year. Defining the cell with the largest suitability index among the 9 connected cells to be the best environment to live in next year, we convert the problem into: for each cell that has fishes $A_{r,c}$, find the best cell around it (including its neighbors and itself). As illustrated in Figure 5, if the best cell is not the cell the fish is staying in, one unit of the fish (50,000 tons) will move to the best cell in the current year. If there are more than one best cell, the fish will pick one best cell to migrate to randomly.

Figure 6. One Unit of Fish Migrate from One Cell to Another



Since the annual change of the ocean temperature is tiny, without loss of generosity, we assume that the distribution of the fish will not make observable change if the observation time is less than one year, and the simulation of 50 times of migration lasts for 50 years. After running the program (see Appendix B) and simulating the process for numerous times, it is found that the predicted locations of the fish in 2070 is stable in some regions, as shown in the following figures.

Figure 7. Distribution of Scottish Herring (left) and Mackerel (right) in 2070



As seen from the above figures, the migration of herring generally displays a pattern of moving north-west rapidly, with a big population gathering at **7.5°W, 66.5°N** and a small portion of the population gathering at **7.5°E, 60°N**.

In contrast, the migration of the mackerel happens a much slower rate. While a large portion of the population moves north-west and gathers at **0°E, 62°N** and **10°W, 62°N**, a small portion of the population moves south-east and gathers at **2°E, 58°N**.

# V Game of Business: Survival of the Small Companies

## 5.1 Small Companies in Danger

Without on-board refrigeration to deliver fresh fish to markets in Scotland fishing ports, the small Scotland-based fishing companies have to navigate back to their bases within a certain period of time, so that the fish they catch will not go bad and have enough left guarantee time to be sold out. Moreover, the largest headway also puts a limit on the navigable distance of the companies,

given that they will continue to operate out of their current locations. Therefore, we have

$$D_m = H_m \times S_m \tag{8}$$

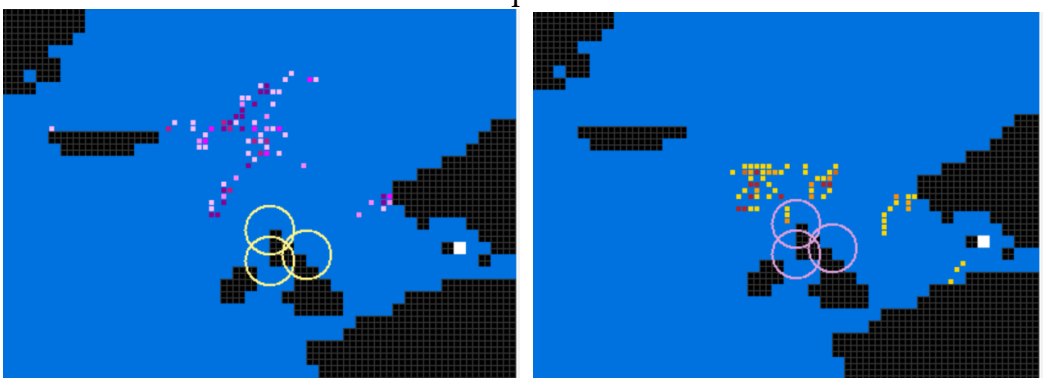
where $H_m$ donates the upper limit of time (hour) to keep the fishes fresh to be eaten raw, $S_m$ donates the habitual headway of a vessel, and $D_m$ donates the largest distance a vessel can travel to keep the caught fish fresh.

According to experiential data, set $H_m$ equals to *3 hours* (to reserve time for selling) and $S_m$ equals to *40 n miles per hour* (that is, around *74 km/h*), which gives $D_m$ equals to *222 km*.

Since one unit difference of longitude represents *111km* and one unit difference of latitude represents $111\times \cos j$ *km*, where *j* is around *4°W*, the biggest distance between the fish habitat and the shore of Scotland can be 2° in longitude, i.e., 4 cells in graph.

In the simulation, we draw three circles centered at three ports that are on the different sides of Scotland, which are respectively Orkney, Stornoway and Eyemouth. In Figure 7, each circle represents the furthest voyage of the small Scotland-based fishing companies, and the phenomenon that no fish stays inside circle, means that the companies reside near that circle do not have enough fish to catch.

Figure 8. Distribution of the Fish in 2070 compared to the Voyage of Small Companies



As discussed in 4.1, (1) gives the difference of temperature $\Delta T$ between a given year $t$ and the year of 2020 that has the biggest probability, and the lower 95% confidence interval and the upper 95% confidence interval produces (2) and (3), which can be used to calculate the smallest and the largest temperature difference. Applying these functions to calculate $\Delta T$, we consider that (1) gives us the most accurate $\Delta T$, (2) gives us the smallest $\Delta T$ (best case for the fishing companies) and (3) gives us the fastest $\Delta T$ (worst case for the fishing companies).

Applying (1), (2), and (3) to our program respectively, the process of the

migration is captured. For the best case, the mostly likely case as well as the worst case, we note the years the herring and the mackerel take for all the in-circle fish to migrate out of the circles. The snapshots of the process of migration is shown in **Appendix A**, and the conclusion is listed here in Table 2.

Table 2. Years for the Fish to be out-circle

| type of fish | best case | most likely case | worst case |
|---|---|---|---|
| Herring | 20 years | 18 years | 15 years |
| Mackerel | 80 years | 61 years | 55 years |

As seen from the above table, it is probably that the small Scotland-based companies cannot catch much herring in 18 years and cannot catch much herring mackerel in 61 years. Even for the best case, it only takes 20 years for the herring and only 80 years for the mackerel to be too far away from the small fishing companies to harvest if the companies continue to operate out of their current locations.

There are some voices saying that since the worst case is calculated to be 15 years and 55 years, no emergent action is needed to take. However, we raise that though the moment of the last unit of fish leave the circle probably will not happen within 15 years, the quantity of the herring and mackerel available for fishing will be continuous declining. Actually, before the "breakpoint" of "too far to harvest" happens, the small fishing companies might have gone broke. Hence, we believe that it is emergent for the small companies to make changes to their operations.

## 5.2 Small Company's Payoff Model and Strategic Adjustments

### 5.2.1 Small Company's Operation model

In 5.1 it has been shown that both Scottish herring and mackerel will migrate too far away for small fishery companies to harvest in a few years. In order to better understand the profit pattern of the small fishing companies, we introduce the small company's operation model in this section.

According to *Scottish sea fisheries statistics 2018* [6], we obtain the total tonnage and value of landings of Scottish herring and mackerel harvested by all Scottish vessels from 2014 to 2018. The average number of herring and mackerel

harvested per year, the average amount of value earned per year and the average value per tonnage are calculated and listed in Table3.

Table 3. the total tonnage and value of landings of Scottish herring and mackerel harvested by all Scottish vessels from 2014 to 2018

| | herring | | | mackerel | | |
|---|---|---|---|---|---|---|
| year | tonnage | value | value / tonnage | tonnage | value | value / tonnage |
| 2014 | 61366 | 18470 | 0.301 | 239470 | 194798 | 0.813 |
| 2015 | 58726 | 21307 | 0.363 | 199887 | 130513 | 0.653 |
| 2016 | 65543 | 43559 | 0.665 | 88487 | 168602 | 1.905 |
| 2017 | 56423 | 23776 | 0.421 | 179956 | 161892 | 0.9 |
| 2018 | 64622 | 24377 | 0.377 | 153000 | 163699 | 1.07 |
| average | 61336 | 26297.8 | 0.429 | 172160 | 163900.8 | 0.952 |
| unit | tonnage | £,000 | £,000 | tonnage | £,000 | £,000 |

From the previous model, we know that the total number of index of Scottish herring is 280 and in current situation, all small fishery companies can harvest 10 index of Scottish herring in total. While the total number of index of mackerel is 158, with 52 index of them caught by small companies. Since the actual average number of Scottish herring harvested per year is 61336 tonnage, the unit of the index is calculated to be $61336 \div 280 \approx 220\ tonnage/index$.

Similarly, we obtain the unit of the index of mackerel to be $172160 \div 158 \approx 1100\ tonnage/index$.

Given that the proportion of herring caught by small fishery companies is 1/28, we know that the total income of all small fishery companies obtained from harvesting Scottish herring is $26297.8 \times 1 \div 28 \approx 939.2\ £,000 = 939{,}200\ £$.

While the total income obtained from mackerel is $163900.8 \times 52 \div 158 = 53{,}942.035\ £,000 = 53{,}942{,}035\ £ \approx 53{,}900{,}000\ £$. The total income of all small fishery companies gained from Scottish herring and mackerel is roughly 55M £(1M=1,000,000).

In particular, if we look at the companies located at east side of Scotland, it is shown from our previous model that even at the year of 2020, only 7 index of mackerel are harvested in total with no income from catching Scottish herring. This means that the total earning of those east-side-located companies is roughly only 7.3M £.

According to the definition of small sized business in UK, a company is regarded as small sized business if its annual turnover is above €2M but below €10M [7]. Therefore, we assume that currently, there are only two companies in each of the fishery area and each company owns two vessels. In this case, if we denote the total number of Scottish herring caught by small fishery companies to

be $x$ tonnage, and the total number of mackerel caught by small fishery companies to be $y$ tonnage, each vessel will harvest $0.25x$ tonnage of Scottish herring and $0.25y$ tonnage of mackerel.

According to Table 3, the average price of Scottish herring is 429 £/tonnage and the average price of mackerel is 952 £/tonnage. Therefore, the total earning $E$ made by harvesting these two species is

$$E = 429 \times 2 \times 0.25x + 952 \times 2 \times 0.25y = 214.5x + 476y \tag{9}$$

If we denote the index of harvested Scottish herring by $X$ and the index of harvested mackerel by $Y$, the relationship between $x$ and $X$ is $x = 220X$. Similarly, the relationship between $y$ and $Y$ is $y = 1100Y$.

Thus, the total earning $E$ made by harvesting these two species becomes

$$E = 214.5x + 476y = 47190X + 523600Y \tag{10}$$

Meanwhile, if we assume that the cost for running one vessel is 1M £ per year, including power cost, labor cost etc., the final profit $F$ made by each small fishery company per year is

$$F = E - 2 \times 1000000 = 47190X + 523600Y - 2000000 \tag{11}$$

### 5.2.2 Attractive Strategies for Small Fishing Companies to Deal with the Migration of Scottish Herring and Mackerel

Given that both Scottish herring and mackerel are migrating further away from their current locations, if small fishery companies do not make adjustments to this phenomenon, less and less fish would be harvested and eventually, all small fishery companies would face a danger of going bankrupt. Thus, we propose two possible strategies for these companies to deal with the migration situation and provide the corresponding new operation model for each of them. One is to relocate the fishing company's assets from the current location a new place which both fish populations are moving to, and the other is to equip the current vessel with cold storage facilities.

Assuming that the cost for relocating the company is 10M £ and the company cannot harvest in the year it relocates, then without buying old storage facilities for the vessels, the profit made by each small fishery company per year is still $F = 47190X + 523600Y - 2000000$. However, the total profit earned in n years $F_1$ is

$$F_1 = \sum_{i=1}^{n}(47190X_1 + 523600Y_1) - 2000000n - 10000000 \tag{12}$$

Meanwhile, since the company relocates at ports closer to where both fish populations are moving, the number of fish caught per year ($X$ and $Y$) will increase significantly, making the profit larger.

If the company choose to equip the current vessel with cold storage facilities, we assume that the cost for buying new facilities to be 1M £ per vessel and the new cost for running one vessel equipped with cold storage facilities is 1.5M £ per year. In this case, the profit made by each small fishery company per year will be

$$F' = 47190X + 523600Y - 3000000 \tag{13}$$

And the total profit earned in n years $F_2$ will be

$$F_2 = \sum_{i=1}^{n}(47190X_2 + 523600Y_2) - 3000000n \tag{14}$$

The benefit of this strategy is that after equipping with cold storage facilities, each vessel can voyage 3 times further than the current maximal voyage distance, which makes it capable of harvesting the fish migrating away.

## 5.3 When some Fisheries migrate abroad

According to *United Nations Convention on the Law of the Sea* [8], foreign vessels are not allowed to fish in territorial waters of another country, if some proportion of the fishery moves into such area, our proposal will need to be adjusted.

From our model, though it is found that all the mackerel will remain in high seas or Scottish territorial sea in 2050, about 7% of Scottish herring will indeed move to territorial waters of Norway by then. Moreover, since the optimal ocean temperature for herring is 5℃, according to our model, most of the herring and mackerel will continue migrating north away from Scotland even after the year of 2050. That is to say, as time goes by, both the herring and mackerel population will eventually end up into the territorial waters of other countries, including Norway, Iceland, Denmark and Greenland.

We observe that the 7% of Scottish herring which end up in territorial waters of Norway will reduce the amount of fish harvested by small fish companies that locates at the east side of Scotland, meaning that even if the companies equip their vessels with cold storage facilities, it does not improve their harvest that much. Thus, a better idea for those companies at east side is to relocate their assets to the north or north-east of Scotland, or to change the target species of harvesting and explore some other sea resources.

Therefore, faced with the loss of fishery when some portion of it moves into the territorial sea of other countries, the most important strategy might not be

investing on cold storage equipment or vessels, but to migrate to the ports where both species are moving to or to seek other resources from the ocean. While in terms of long-distance oceangoing voyage, the refrigeration is the key. Hence, if some proportion of the fishery moves into the territorial waters of another country, our proposal needs to be adjusted accordingly.

# VI Model Evaluation and Further Discussion

## 6.1 Evaluation of Models

To explore the future location of herring and mackerel, we establish two models based on a large amount of data, least square regression, and normal distribution. To explore the future situation of Scottish small fishing companies, we further establish the Small Company's Payoff Model. We answer the questions convincingly and vividly by using multiple pictures of our simulation result.

Throughout our modeling, we collect a large amount of reliable data to make our prediction and stimulation result precisely. Instead of only finding the location with the most suitable temperature, we also stimulate the traveling path of the 2 kinds of fishes respectively. To better present our simulation results, we use many colorful pictures to show the change of ocean temperature, the change of fish distribution and the number of fishes in the area that small fishing companies can harvest.

Moreover, our Small Company's Payoff Model takes many economic factors into accounts, such as the cost of relocating and purchasing equipment, the minimum incomes that prevent a small company from bankruptcy, and the haul of each vessel. To figure out the effectiveness of our proposal, we test these operations in our simulated future situation. The simulation shows a better profit for a small fishing company, which verifies the rationality of our proposal.

However, our models still have the following weaknesses:

First, although we calculated the suitability index based on reliable predicted temperature, we didn't consider other environmental factors such as the swam behavior of fishes, the amount of plankton and the salinity in different areas.

Second, since we cannot find reliable data of Scottish small company, we assume the values of some factors based on the general situation.

## 6.2 Further Discussion

Our paper establishes three models, including the Ocean Warming Model, the Habitat Suitability Evaluation Model, and the Small Company's Payoff Model. We use Ocean Warming Model to generate the temperature at each point and used Habitat Suitability Evaluation Model to determine the corresponding suitability index. We simulate the annual change of current habitat based on the suitability index and get the annual location of these fishes. Then, we establish the Small Company's Payoff Model to figure out the future situation of Scottish small fishing companies and propose operations for them. Finally, we consider the territory water of other countries and further improve our proposal.

As mentioned in 8.1, there are two weaknesses in our model that can be improved. To make the Habitat Suitability Evaluation Model more accurate, we need to collect data about other related factors and to figure out the relation between them. To improve the Small Company's Payoff Model, we need to find information about the economic situation in Scotland, such as land price, equipment price, and labor cost.

**Magazine Article**

# Saving Scotland-based Fishing Companies in Danger: Go North as the Fishes Do

Scotland-based fishing companies are now facing a severe situation where there would be no enough fish to harvest. Two main species of Scottish fishery harvest, Scottish herring, and mackerel, are moving north away from the main coast of Scotland due to the increasing ocean temperature, seriously influencing the earning of Scotland-based fishing companies, small companies in particular. Practical and economical strategies need to be taken to protect and promote the development of these companies.

Global warming has always been a serious issue addressed and discussed by various countries and organizations. The rising temperature is creating a huge impact on every aspect of human and other creature's life. One typical example is the unusual migration of animals. When temperature changes are too great for maintaining the quality of habitats for certain species and threaten their continued thriving, the species will move to seek other habitats which are more suitable for their present and future living and successful reproduction. Without doubt, such things are happening on Scottish herring and mackerel.

Scottish herring and mackerel are moving north away from their current habitats around Scotland. Researchers have found that the most preferable living temperature for Scottish herring in January is 5℃, while for mackerel, the temperature preference is 10.2 ℃. The suitable ocean temperature around Scotland used to make it the ideal habitat for both species, and for long, Scottish herring and mackerel have brought huge economic profits for Scotland. However, in recent years, the increasing sea temperature is forcing the two species to migrate north away from Scotland.

Based on the data of the average global sea surface temperature (SST) from 1950 to 2015, our group simulated the global SST change in the next 30 years and were sad to find out that both Scottish herring and mackerel are moving north away from Scotland very fast, especially for Scottish herring. In fact, they are migrating so fast that within only a few years, their habitats will be so far away that small fishery companies cannot even catch them anymore!

To be more specific, out group simulated the best case, worst case, and most likely case and estimated the elapsed time of each case when populations of the two species will be too far away for small fishing companies to harvest if the small fishing companies do not take any action and remain at their current locations. The result turned out to be horrifying. The most likely elapsed time for Scottish

herring to be too far away to be caught is 18 years, even in the best case, the elapsed time is only 20 years. Let along the worst case where it just takes 15 years for all Scottish herring to move too far away. As for mackerel, things seem to be a little better. It takes 61 years for mackerel to be out of reach by small fishery companies in the most likely case, 55 years in the worst case and 80 years in the best. However, what needs to be pointed out here is that we calculated the time when NO fish could be harvested. And for a company to operate and survive, however, it needs to catch ENOUGH fish rather than just catch one fish to avoid going bankrupt. Therefore, it is indeed urgent to take action.

Based on our research, our group proposes three possible strategies to deal with the current situation. According to our findings, most of the Scottish herring and mackerel are moving north-west of Scotland. So our first proposal for a fishing company is to relocate some or all of its assets to the north or north-west port of Scotland, in which case more fish can be harvested. Another possible solution is to equip the current vessel with cold storage facilities so as to enlarge the time a raw fish can keep fresh. This makes it possible for the vessel to have more time to voyage to longer distance and catch more fish, thus making more profits for the company. The last strategy is to encourage cooperation between small fishery companies and large fishery companies. Given that usually, small fishery companies only have limited or very limited financial resources to invest in new equipment or vessels, two or more companies can raise money together to buy new equipment or vessels and fairly share the profits gain from the investment. Or small fishery companies can also rent other company's equipment or vessel. No matter in which case, fishery companies can more precisely deal with the migration of Scottish herring and mackerel.

Although under the influence of global warming, Scotland-based fishing companies are in danger of not harvesting enough fish due to the migration of Scottish herring and mackerel, they still have time to take an efficient measurement and make adjustments to the current situation. As long as suitable practical and economical strategies are taken, the development of Scotland-based fishing companies can be protected and promote for sure, voyaging to a brighter future.

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[7] Ezez. 2020. "Finding a standard definition for Small Businesses". Towergate Insurance. https://www.towergateinsurance.co.uk/business-insurance/finding-a-standard-definition-for-small-businesses.

[8] Ezez. 2020. "Overview - Convention & Related Agreements". Un.org. https://www.un.org/depts/los/convention_agreements/convention_overview_convention.htm

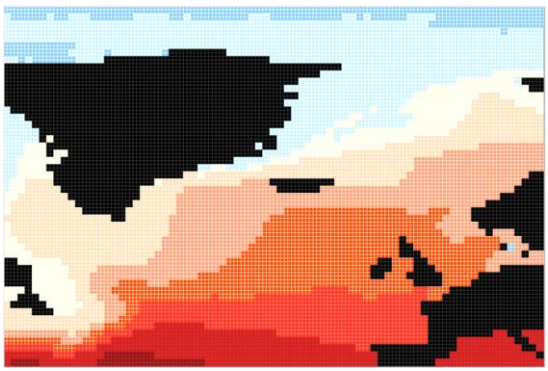
# Appendixes

## Appendix A: Figures

**For Problem 1:**

Temperature from 2020 to 2070 (per 10 years)

2020

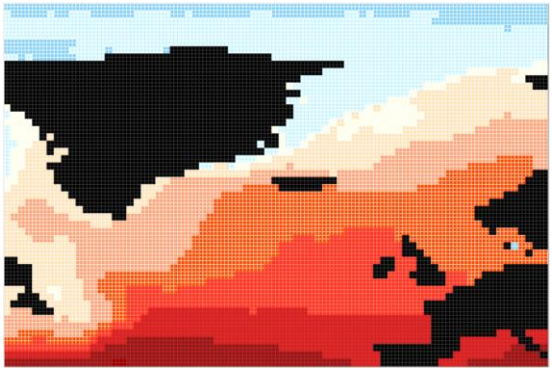


2030

2040

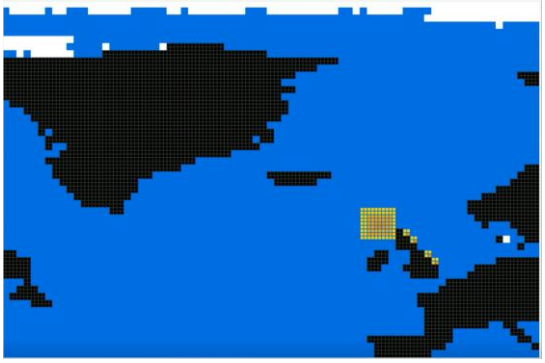
2050

2060

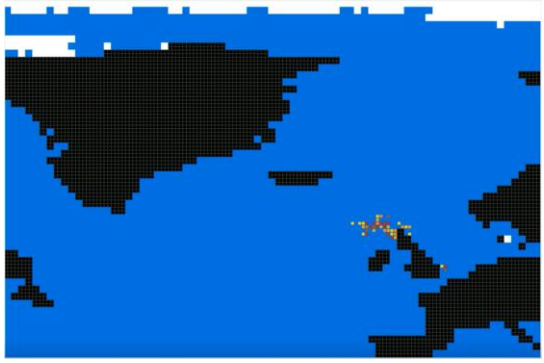


2070

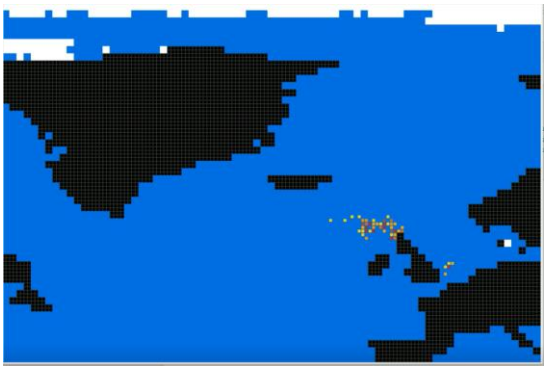
Distribution of Mackerel 2020 to 2070 (per 10 years)



2020



2030



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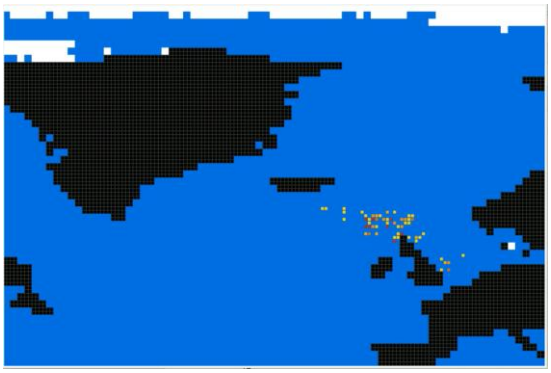
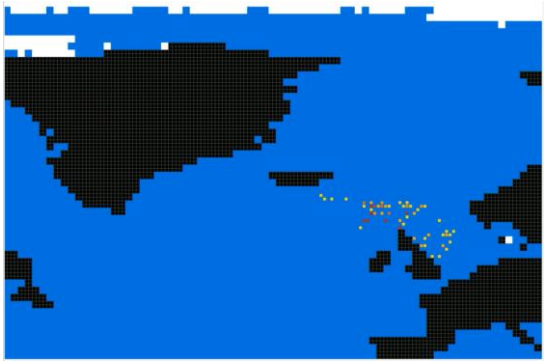
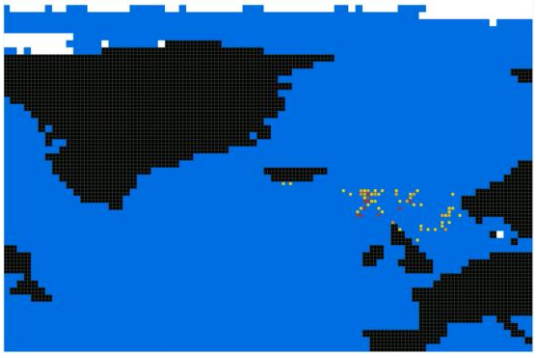


2050

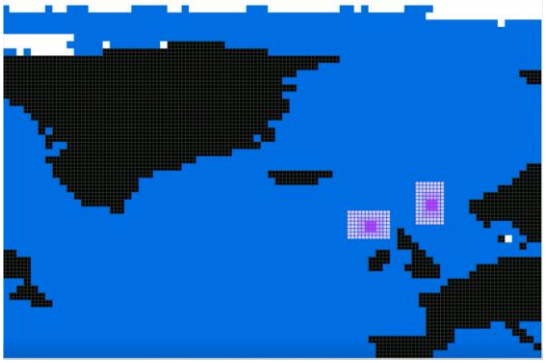


2060

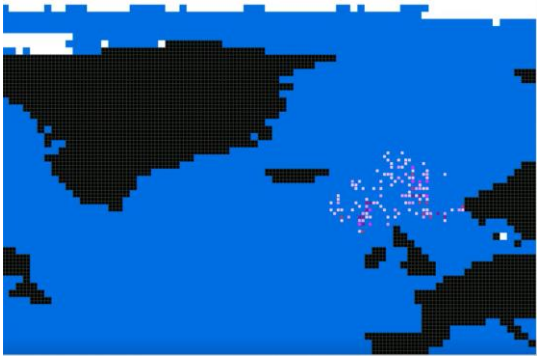


2070

Distribution of Herring 2020 to 2070 (per 10 years)



2020



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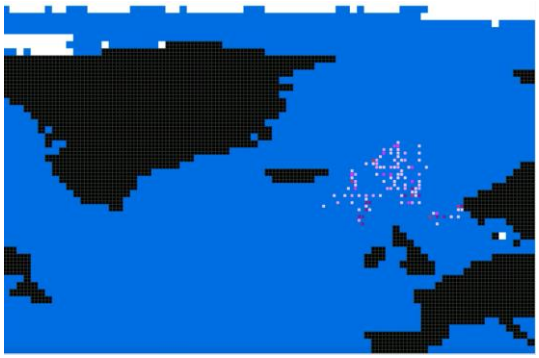

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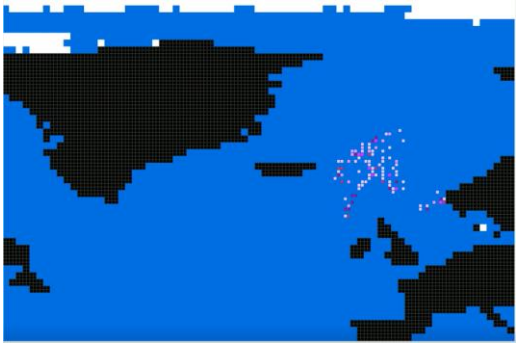

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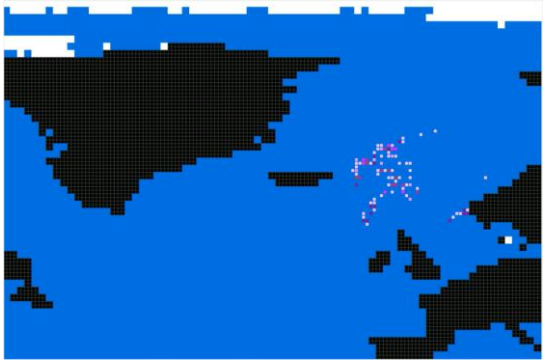

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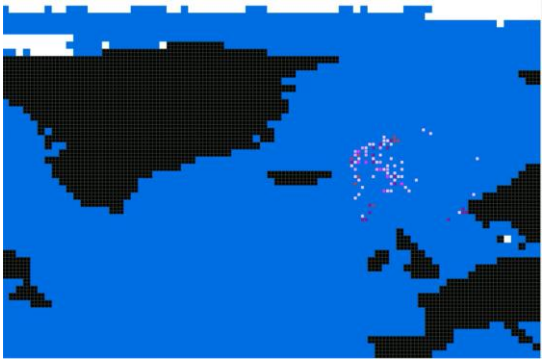
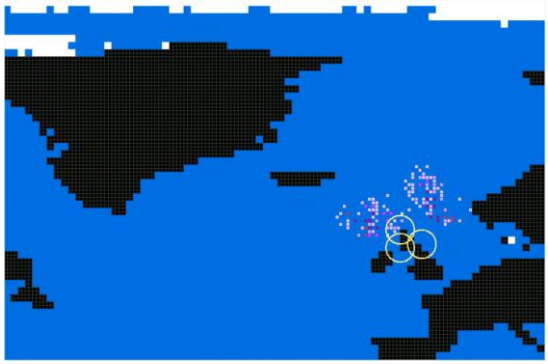


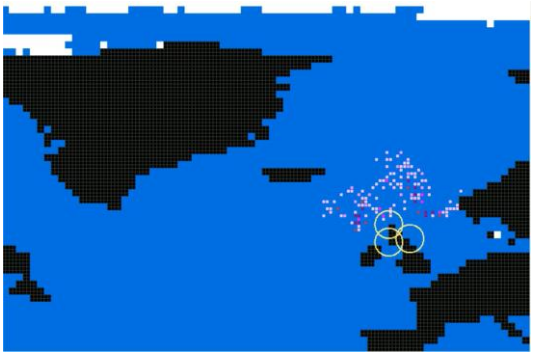
2070

**For Problem 2**

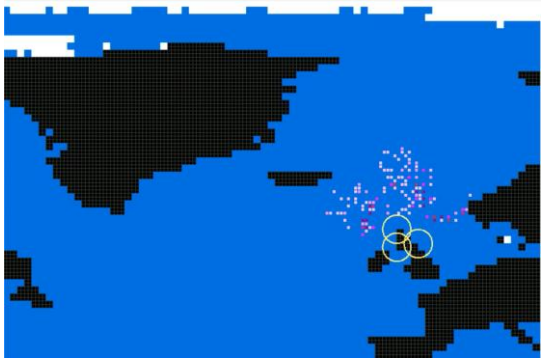
Herring Worst Case:



2025

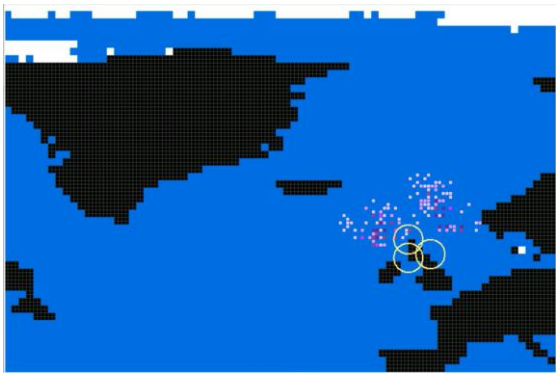


2030

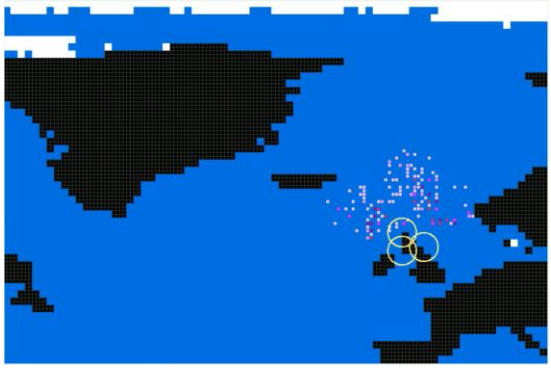


2035

Herring Most Likely Case:



2026



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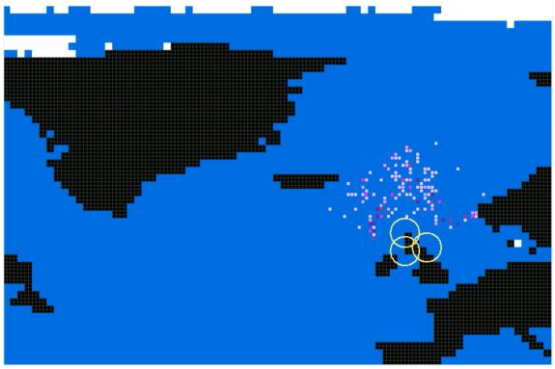
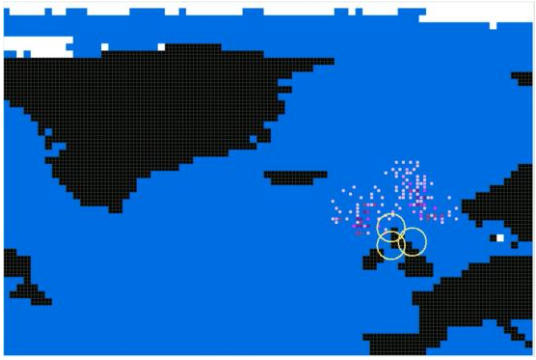


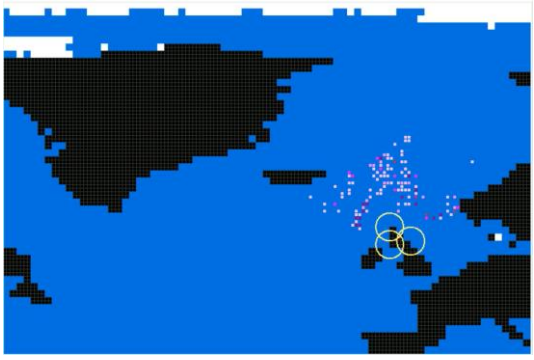
2038

Herring Best Case:



2027

2034



2040

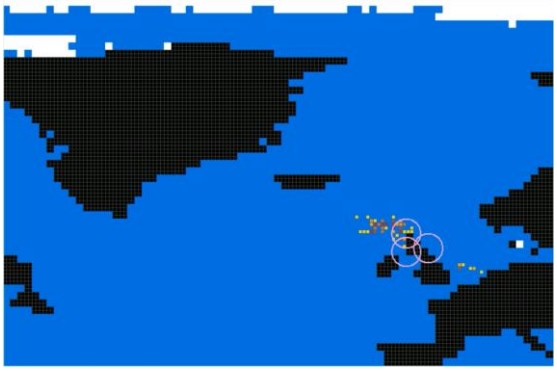
Mackerel Worst Case:

2038

2056

2077

Mackerel Most Likely Case:



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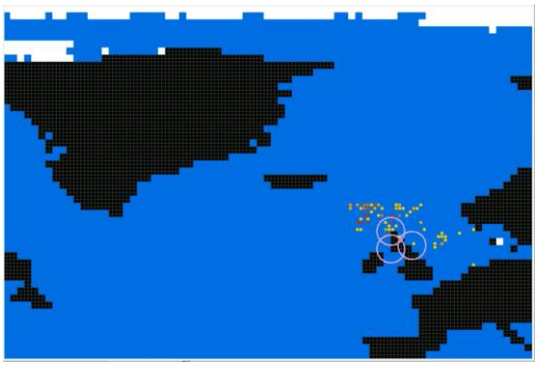

2060

2081

Mackerel Best Case:

2047

2074

2100

# Appendix B: Code

MATLAB Source Code: Least Square Regression

```
function a = LSR(x,y,n,lambda)
A=[];
B=[y;zeros(n+1,1)];
for i=0:n
    A(:,i+1)=x.^i;
end
A=[A;eye(n+1).*sqrt(lambda)];
A_T=A.';
a=flipud((A_T*A)\(A_T*B));
end
```

MATLAB Source Code: Calculate the coefficients of Temperature Model

```
format long
A=xlsread('jan.csv');
x=A(:,1);
y1=A(:,2);
y2=A(:,3);
y3=A(:,4);
close all;
training_error0=1;
n=4;
a1=LSR(x,y1,n,1);
a2=LSR(x,y2,n,1);
a3=LSR(x,y3,n,1);
disp("Coefficients: ")
disp('a1 :')
disp(a1);
disp('a2 :')
disp(a2);
disp('a3 :')
disp(a3);
```

MATLAB Source Code: plot the Annual anomaly in 2020 and the Predicted value in 2070

```
plot(x,y1,'.'); hold on;
plot(x,y2,'.');
plot(x,y3,'.');
disp(polyval(a1,2020));
disp(polyval(a2,2020));
disp(polyval(a3,2020));
x=linspace(1940,2080,100);
plot(x,polyval(a1,x));
plot(x,polyval(a2,x));
plot(x,polyval(a3,x));
```

Python Source Code: Temperature Prediction based on Temperature Model

```
def temperature(year,h):
    if h=='m':
        delta_T=0.000000000040393*year**4-0.000000154772254*year**3+...
        0.000146765566744*year**2+0.002578002667371*year+0.000005206079179- 0.907903496194830
    if h=='l':
        delta_T=0.000000000003363*year**4-0.000000007835048*year**3+...
        0.000001363526822*year**2+0.001755275153093*year+0.000003547457095- 0.527997611431180
    if h=='u':
        delta_T=0.000000000077423*year**4-0.000000301709461*year**3+...
        0.000292167606182*year**2+0.003400730473337*year+0.000006864701920- 1.287809380911568
    tem=[]
    for i in range(100):
        d=[]
        for j in range(150):
            T=data1[i][j]
            if T==-1000 or T==32768:
                d.append(T)
            else:
                T_change=T/100+delta_T
                d.append(T_change)
        tem.append(d)
    return tem
```

Python Source Code: Calculation of Suability Index for Mackerel

```
def tem_cond(t):
    if t==-1000 or t==32768:
        return -99999

    f=(1/(2.5*np.sqrt(2*np.pi)))*np.exp(-np.square(t-10.2)/(2*2.5*2.5))

    return 6.25*f
```

Python Source Code: Calculation of Suability Index for Herring

```
def tem_cond(t):
    if t==-1000 or t==32768:
        return -99999

    f=(1/(2.5*np.sqrt(2*np.pi)))*np.exp(-np.square(t-5)/(2*2.5*2.5))

    return 6.25*f
```

Python Source Code: Find the Best Box

```
def choose_best(r,c,temp_grid):
    best_r=[r]
    best_c=[c]
    best_t=tem_cond(temp_grid[r][c])
    for i in [r-1,r,r+1]:
        for j in [c-1,c,c+1]:
            if tem_cond(temp_grid[i][j])>best_t:
                best_t=tem_cond(temp_grid[i][j])
                best_r.clear()
                best_c.clear()
                best_r.append(i)
                best_c.append(j)
            elif tem_cond(temp_grid[i][j])==best_t:
                best_r.append(i)
                best_c.append(j)
    n=np.random.randint(len(best_r))
    return [best_r[n],best_c[n]]
```

Python Source Code: Use different colors to represent different types of boxes

```
def make_disp_grid(canvas,fishMap):
    rows = len(fishMap)
    cols = len(fishMap[0])
    cell_width = int(canvas.cget('width')) / cols
    cell_height = int(canvas.cget('height')) / rows

    disp_grid = [['' for col in range(cols)] for row in range(rows)]
    for row in range(rows):
        for col in range(cols):
            x1 = col * cell_width
            y1 = row * cell_height
            x2 = x1 + cell_width
            y2 = y1 + cell_height

            w=0.01
            c='#0072e0'
            if int(fishMap[row][col])==-1:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='white',width=w,outline='white')
            elif int(fishMap[row][col])==-2:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='black',width=w,outline='#363636')
            elif int(fishMap[row][col])==0:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#0072e0',width=w,outline=c)
            elif int(fishMap[row][col])==1:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#FFD700',width=w,outline=c)
            elif int(fishMap[row][col])==2:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#FFA500',width=w,outline=c)
            elif int(fishMap[row][col])==3:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#FF8C00',width=w,outline=c)
            elif int(fishMap[row][col])==4:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#D2691E',width=w,outline=c)
            elif int(fishMap[row][col])==5:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#A0522D',width=w,outline=c)
            elif int(fishMap[row][col])>5:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#B22222',width=w,outline=c)
    return disp_grid
```

Python Source Code: All fishes go to the Best Box around them

```
def update(grid,year,h):
    temp_grid=temperature(year,h)
    rows = len(grid)
    cols = len(grid[0])
    new_grid=grid.copy()
    for row in range(rows):
        for col in range(cols):
            if grid[row][col]>0:

                row1,col1=choose_best(row,col,temp_grid)
                if grid[row1][col1]<0:
                    continue
                elif row!=row1 or col!=col1:

                    new_grid[row][col]-=1
                    new_grid[row1][col1]+=1
```

Python Source Code: Display the annual distribution of fishes

```
def display(canvas, Map, disp_grid):
    for row in range(len(grid)):
        for col in range(len(grid[0])):
            w=0.01
            c='#0072e0'
            if int(Map[row][col])==0:
                canvas.itemconfig(disp_grid[row][col],fill='#0072e0',width=w,outline=c)
            elif int(Map[row][col])==1:
                canvas.itemconfig(disp_grid[row][col],fill='#FFBBFF',width=w,outline=c)
            elif int(Map[row][col])==2:
                canvas.itemconfig(disp_grid[row][col],fill='#FF83FA',width=w,outline=c)
            elif int(Map[row][col])==3:
                canvas.itemconfig(disp_grid[row][col],fill='#FF00FF',width=w,outline=c)
            elif int(Map[row][col])==4:
                canvas.itemconfig(disp_grid[row][col],fill='#C71585',width=w,outline=c)
            elif int(Map[row][col])==5:
                canvas.itemconfig(disp_grid[row][col],fill='#B03060',width=w,outline=c)
            elif int(Map[row][col])>5:
                canvas.itemconfig(disp_grid[row][col],fill='#8B008B',width=w,outline=c)
    canvas.update()
```

Python Source Code: Initialize the Distribution of fishes

```
def populate(leftTop,RightBottom):
    width=RightBottom[1]-leftTop[1]
    height=RightBottom[0]-leftTop[0]
    mean=((leftTop[0]+RightBottom[0])/2,(leftTop[1]+RightBottom[1])/2)

    for r in range(leftTop[0],RightBottom[0]):
        for c in range(leftTop[1],RightBottom[1]):
            if fishMap[r][c]==0:
                count=1
                if abs(c-mean[1])<width/3 and abs(r-mean[0])<height/3:
                    count+=1
                if abs(c-mean[1])<width/6 and abs(r-mean[0])<height/6:
                    count+=1
                fishMap[r][c]=count
```

Python Source Code: Build up the Background based on temperature data

```
Year = open('2020.txt', 'r')
store=Year

data0 = []
rowth=0
for line in store:
    rowth+=1
    if rowth!=1:
        line.strip('\n')
        change=line.replace('-32768',' 32768')
        row=change.split()
        data0.append(row)
Year.close()

data1=[]
for i in range(0,50):
    c=[]
    for j in range(120,195):
        c.append(int(data0[i][j]))
        c.append(int(data0[i][j]))
    data1.append(c)
    data1.append(c)

fishMap=disp_grid = [[0 for col in range(150)] for row in range(100)]
for r in range(len(data1)):
    for c in range(len(data1[0])):
        if int(data1[r][c])==-1000:
            fishMap[r][c]=-1
        elif int(data1[r][c])==32768:
            fishMap[r][c]=-2
```

Python Source Code: Show the Results

```
from tkinter import *
from PIL import Image
import numpy as np
from time import sleep

window = Tk()
window.title('Map')

f = Frame(window)
f.pack()

canvas = Canvas(f, bg='white', width=750, height=500,bd=0.1)
canvas.pack()

year=0
total_year=0
grid=fishMap.copy()
dg=make_disp_grid(canvas,grid)
while year<=50:

    display(canvas,grid,dg)
    sleep(0.1)
    update(grid,year+2020,'m')
    # 'm' : mean value, 'l': lower 95% value, 'u': upper 95% value
    year=year+1
    oval1 = canvas.create_oval(530,300,570,340,outline='#FFF68F',width=1.5)
    oval2 = canvas.create_oval(530,325,570,365,outline='#FFF68F',width=1.5)
    oval3 = canvas.create_oval(500,260,660,420,outline='#FFF68F',width=1.5)

    print(year)
canvas.mainloop()
```

Python Source Code: Display the Temperature Distribution of each year

```
window = Tk()
window.title('Map')
f = Frame(window)
f.pack()
canvas = Canvas(f, bg='gray', width=750, height=500,bd=0.1)
canvas.pack()

def make_disp_grid(canvas, rows, cols):
    cell_width = int(canvas.cget('width')) / cols
    cell_height = int(canvas.cget('height')) / rows
    disp_grid = [['' for col in range(cols)] for row in range(rows)]
    for row in range(rows):
        for col in range(cols):
            x1 = col * cell_width
            y1 = row * cell_height
            x2 = x1 + cell_width
            y2 = y1 + cell_height
            if int(data2[row][col])==-1000:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#87CEFA',outline='#F0FFFF')
            elif int(data2[row][col])==32768:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='black',outline='#363636')
            elif int(data2[row][col])>=21:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#8B0000',outline='#CD2626')
            elif 21>int(data2[row][col])>=18:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#8B1A1A',outline='#CD2626')
            elif 18>int(data2[row][col])>=15:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#CD2626',outline='#EE2C2C')
            elif 15>int(data2[row][col])>=12:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#FF3030',outline='#FF7F50')
            elif 12>int(data2[row][col])>=9:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#FF4500',outline='#FFDAB9')
            elif 9>int(data2[row][col])>=6:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#FFA07A',outline='#FFEFDB')
            elif 6>int(data2[row][col])>=3:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#FFE4B5',outline='#FFFAFA')
            elif 3>int(data2[row][col])>0:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#FFFFF0',outline='#FFFFF0')
            elif int(data2[row][col])<=0:
                disp_grid[row][col] = canvas.create_rectangle(x1,y1,x2,y2,fill='#F0FFFF',outline='#BFEFFF')

    return disp_grid

make_disp_grid(canvas, 100, 150)
window.mainloop()
```